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Studierichting: Informatica
Oefeningenreeks 6E nr. 1.3

Bepaal de Taylorpolynoom $T_5(f, 0)(x)$ voor $f(x) = e^{\sin x}$ door de afgeleiden te berekenen in $x = 0$.

$$f(x) = e^{\sin x} \implies f(0) = e^0 = 1$$

$$f'(x) = f(x) \cos x \implies f'(0) = 1 \cdot 1 = 1$$

$$f''(x) = f'(x) \cos x - f(x) \sin x \implies f''(0) = 1 \cdot 1 - 1 \cdot 0 = 1$$

$$f'''(x) = f''(x) \cos x - 2f'(x) \sin x - f(x) \cos x \implies f'''(0) = 1 \cdot 1 - 0 - 1 \cdot 1 = 0$$

$$f^{iv}(x) = f'''(x) \cos x - 3f''(x) \sin x - 3f'(x) \cos x + f(x) \sin x \implies f^{(4)}(0) = 0 - 0 - 3(1)(1) + 0 = -3$$

$$f^v(x) = f^{iv}(x) \cos x - 4f'''(x) \sin x - 6f''(x) \cos x + 4f'(x) \sin x + f(x) \cos x$$

$$\implies f^{(5)}(0) = -3(1) - 0 - 6(1)(1) + 0 + 1(1) = -3 - 6 + 1 = -8$$

$$T_5(x) = 1 + \frac{1}{1}x + \frac{1}{2}x^2 + \frac{0}{6}x^3 + \frac{-3}{24}x^4 + \frac{-8}{120}x^5$$

$$T_5(x) = 1 + x + \frac{1}{2}x^2 - \frac{1}{8}x^4 - \frac{1}{15}x^5$$

References